

Examiner
only

- (i) Use letters **N** and **n** to complete the Punnett square to show how parents **1** and **2** could produce offspring number **4**. [2]

Gametes		

- (ii) State the genotype of individual **9**. Explain your answer. [2]

.....

.....

- (iii) State the possible genotypes of individual **5**. [1]

.....

- (c) Our understanding of how characteristics are inherited comes from the principles proposed by Gregor Mendel in the 19th century. In his experiments on pea plants (*Pisum sativum*) Mendel studied characteristics controlled by a single gene. State why this is not representative of most phenotypes. [1]

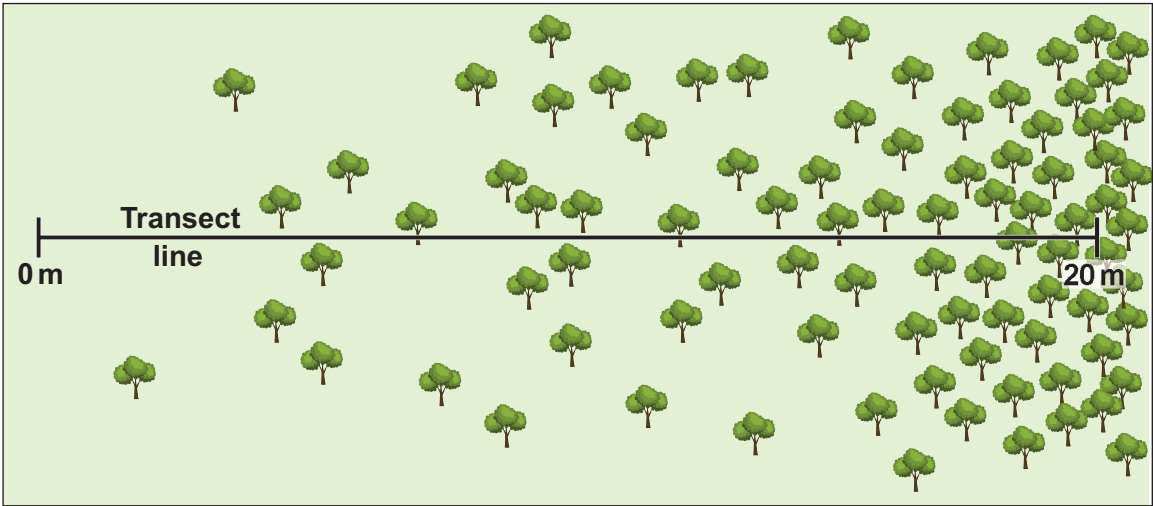
.....

.....

8



4. Rhys wanted to investigate the abundance and distribution of four plant species as he moved from the edge of a woodland towards the centre. He laid down a tape measure to form a transect. Using a 0.25m² quadrat he sampled every five metres along the transect line, estimating the percentage area cover of each species. The diagram shows the area of woodland that was investigated.



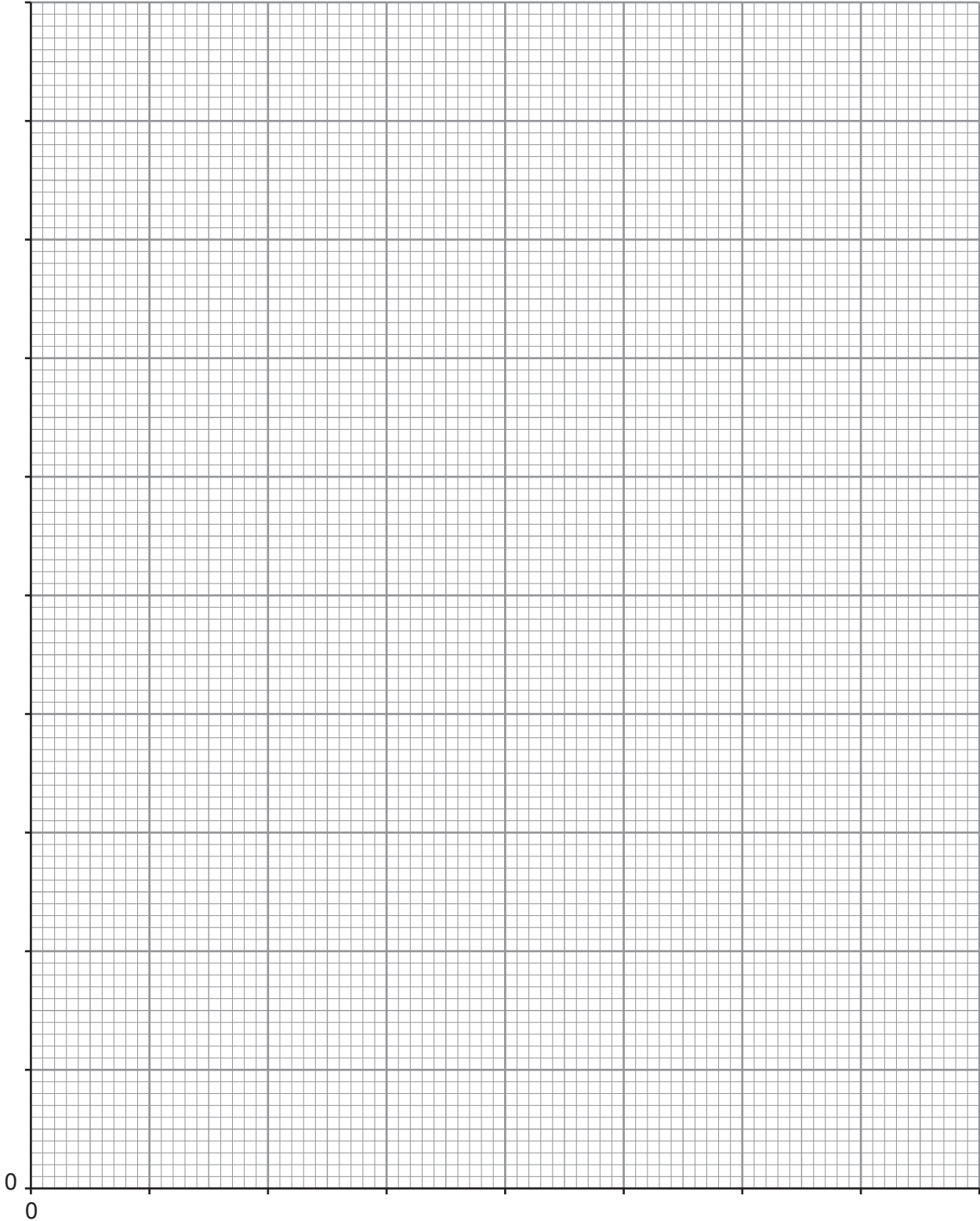
His results are shown in the table.

Species	Percentage cover at each distance along transect (m)				
	0	5	10	15	20
Primrose (<i>Primula vulgaris</i>)	40	35	35	25	5
Dog's Mercury (<i>Mercurialis perennis</i>)	50	35	15	10	5
Bluebell (<i>Hyacinthoides non-scripta</i>)	0	5	10	25	80
Wood Anemone (<i>Anemone nemorosa</i>)	5	20	40	35	10



Examiner
only

(a) **Draw a line graph** of the results for both Dog’s Mercury and Bluebell on the grid below. Label each line. [4]



Examiner
only

(b) (i) State the term used to refer to competition between two different species. [1]

.....

(ii) Describe the results for Dog’s Mercury and Bluebell. Suggest an explanation for your answer. [2]

.....
.....
.....
.....
.....

(c) (i) Jim suggested to Rhys that he should have carried out random sampling. State why it would be unsuitable to use a random sampling method in this investigation. [1]

.....
.....
.....

(ii) Suggest **one** improvement Rhys could make to his method. [1]

.....
.....

(d) Biodiversity is a measure of the variety of different species and the number of individuals within those species in an area. State **two** reasons why it is important that biodiversity is maintained. [2]

.....
.....
.....
.....

